

**ASSIGNMENT No: 8****TITLE: Write a Java program to demonstrate the use of the super keyword in inheritance.**

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Code

```
class Animal{
    String color="White";
    Animal(){
        System.out.println("Animal constructor");
    }

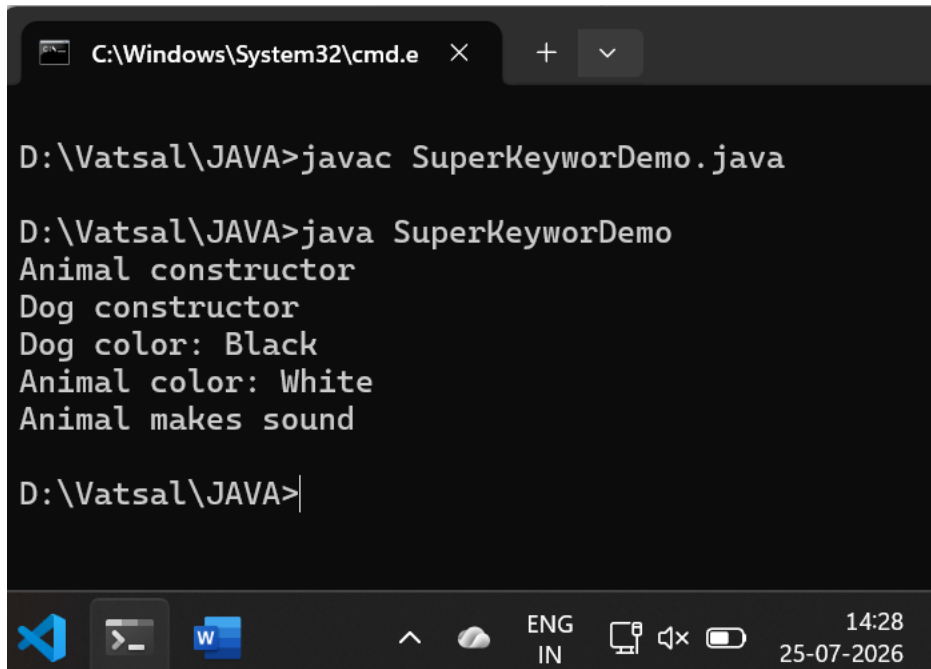
    void sound(){
        System.out.println("Animal makes sound");
    }
}

class Dog extends Animal{
    String color="Black";
    Dog(){
        super();
        System.out.println("Dog constructor");
    }

    void display(){
        System.out.println("Dog color: "+color);
        System.out.println("Animal color: "+super.color);
        super.sound();
    }
}
```

```
}  
  
public class SuperKeyworDemo {  
    public static void main(String[] args) {  
        Dog dog = new Dog();  
        dog.display();  
    }  
}
```

Output



```
C:\Windows\System32\cmd.e  
  
D:\Vatsal\JAVA>javac SuperKeyworDemo.java  
  
D:\Vatsal\JAVA>java SuperKeyworDemo  
Animal constructor  
Dog constructor  
Dog color: Black  
Animal color: White  
Animal makes sound  
  
D:\Vatsal\JAVA>|
```

Exercise

1. Create an **Employee and Manager program** where Manager accesses Employee details using super keyword.

Code

```
class Employee {  
    String name = "Vatsal";  
    double salary = 50000;  
  
    Employee() {  
        System.out.println("Employee profile created");  
    }  
}
```

```

    void displayInfo() {
        System.out.println("Role: General Employee");
    }
}

class Manager extends Employee {
    String name = "Parth";
    double bonus = 15000;

    Manager() {
        super();
    }

    void showDetails() {

        System.out.println("Manager Name: " + name);
        System.out.println("Employee Name: " + super.name);

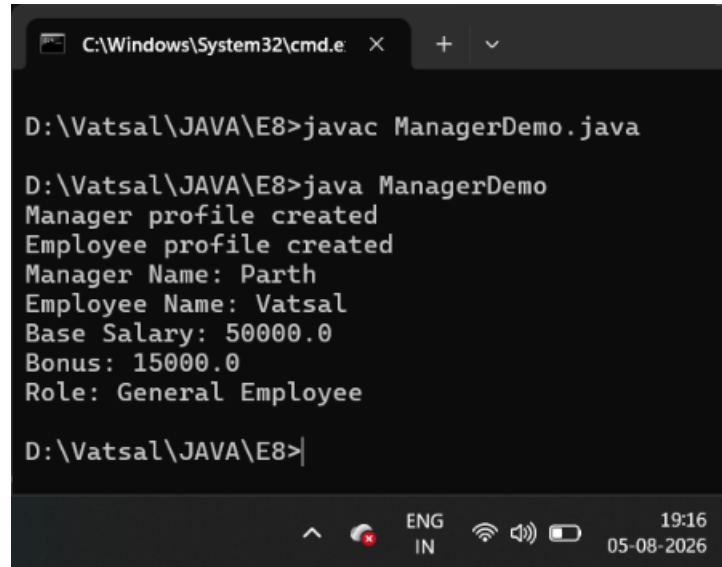
        System.out.println("Base Salary: " + salary);
        System.out.println("Bonus: " + bonus);

        super.displayInfo();
    }
}

public class ManagerDemo {
    public static void main(String[] args) {
        System.out.println("Manager profile created");
        Manager mgr = new Manager();
        mgr.showDetails();
    }
}

```

Output



```
C:\Windows\System32\cmd.e X + v

D:\Vatsal\JAVA\E8>javac ManagerDemo.java

D:\Vatsal\JAVA\E8>java ManagerDemo
Manager profile created
Employee profile created
Manager Name: Parth
Employee Name: Vatsal
Base Salary: 50000.0
Bonus: 15000.0
Role: General Employee

D:\Vatsal\JAVA\E8>
```

2. Develop a **Vehicle Insurance System** where child insurance classes access parent vehicle information using the super keyword.

Code

```
class Vehicle {
    String vehicleType = "Car";
    String model = "Honda City";
    int basePremium = 5000;

    Vehicle() {
        System.out.println("Vehicle registration details fetched.");
    }

    void displayVehicleDetails() {
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Vehicle Model: " + model);
    }
}

class CarInsurance extends Vehicle {
    int basePremium = 7500;
    int addOnCoverage = 1500;

    CarInsurance() {
        super();
        System.out.println("Car Insurance policy generated.");
    }
}
```

```

void calculateAndDisplay() {
    System.out.println("\nVehicle Details");

    super.displayVehicleDetails();

    System.out.println("\nPolicy Details");

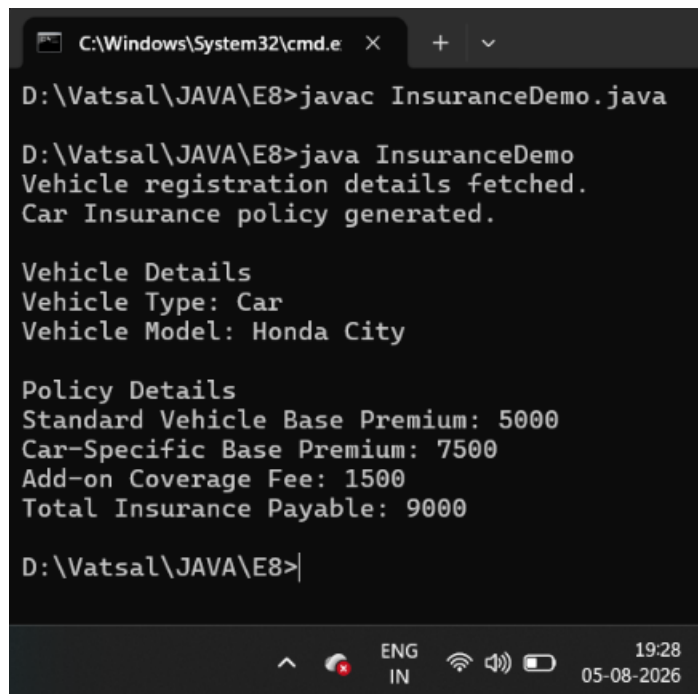
    System.out.println("Standard Vehicle Base Premium: " + super.basePremium);
    System.out.println("Car-Specific Base Premium: " + basePremium);
    System.out.println("Add-on Coverage Fee: " + addOnCoverage);

    int totalInsuranceCost = basePremium + addOnCoverage;
    System.out.println("Total Insurance Payable: " + totalInsuranceCost);
}
}

public class InsuranceDemo {
    public static void main(String[] args) {
        CarInsurance policy = new CarInsurance();
        policy.calculateAndDisplay();
    }
}

```

Output



```

C:\Windows\System32\cmd.e X + v
D:\Vatsal\JAVA\E8>javac InsuranceDemo.java

D:\Vatsal\JAVA\E8>java InsuranceDemo
Vehicle registration details fetched.
Car Insurance policy generated.

Vehicle Details
Vehicle Type: Car
Vehicle Model: Honda City

Policy Details
Standard Vehicle Base Premium: 5000
Car-Specific Base Premium: 7500
Add-on Coverage Fee: 1500
Total Insurance Payable: 9000

D:\Vatsal\JAVA\E8>

```